

Structured Semiconductor Financial News and Volatility Dataset (2020–2025)

A reusable research dataset combining 23,430 institutional news headlines with ETF market data, Garman–Klass volatility, FinBERT sentiment scores, and intraday microstructure variables for SMH and SOXX.

01 Dataset Overview

The first openly documented dataset that aligns institutional semiconductor news with high-frequency ETF microstructure, ready for volatility forecasting and event studies.

23,430	2020–25	1,568 TRADING DAYS	2 ETFs: SMH / SOXX	5 FORECAST HORIZONS
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02 Data Pipeline

Two parallel processing tracks — market microstructure and text — converge into a single analytical forecasting table at daily frequency.

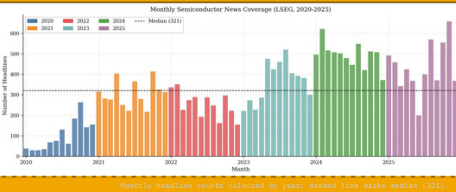


03 Annual News Coverage

ANNUAL SEMICONDUCTOR NEWS COVERAGE (LSEG, 2020–2025)



Coverage roughly 5x from 2020 to 2024, peaking during the AI-driven re-rating of semiconductor equities. Sample slightly contracts in 2025 as the news cycle normalises.



04 Variables

All variables are aligned at the daily trading-day frequency and ticker-keyed for SMH and SOXX.

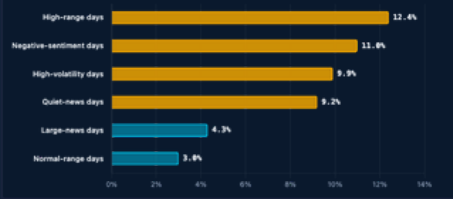
04.01 Headline records	5 fields
date · time · source · ticker · headline	
04.02 Market & microstructure	8 fields
Open · High · Low · Close · Volume · GK_volatility · log_A1_range · rel_volume	
04.03 FinBERT sentiment	2 fields
s_i - headline-level net sentiment · s_t - daily aggregate = mean(p - n)	

05 Key Finding

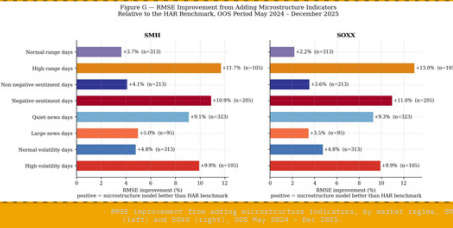
A microstructure-augmented HAR model reduces forecast RMSE by 5.8–9.9% versus the HAR benchmark. Adding FinBERT sentiment contributes a further +0.5–0.9%. Gains concentrate where they matter — high-range and high-volatility days (see improvements of 10–13%).

06 Conditional RMSE Improvement

CONDITIONAL RMSE IMPROVEMENT: MICROSTRUCTURE VS HAR



Improvement is largest where the benchmark is weakest — episodes of elevated intraday range and realised volatility. Effect attenuates on quiet or normal-range days, suggesting the gain is concentrated in regimes practitioners care about most.



RMSE improvement from adding microstructure indicators, by market regime, SMH (left) and SOXX (right), BOS May 2024 - Dec 2025.

07 Use Cases

- Financial NLP research**
Benchmarking FinBERT & LLM sentiment models
- Volatility forecasting**
HAR, GARCH, ML & deep-learning baselines
- Event studies**
Sanctions, export-control & policy shocks
- Semiconductor market research**
SMH / SOXX, sector cycles, capex regimes
- AI regime analysis**
Pre / post-ChatGPT & export-control eras

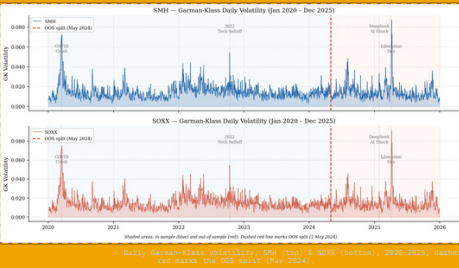
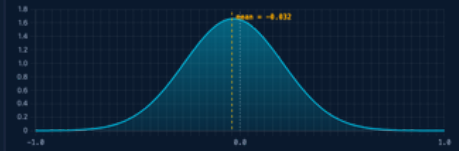
08 Technical Stack

All processing in open-source Python; reproducible from the included Jupyter notebook.

- Python 3.11
- pandas
- numpy
- yfinance
- HuggingFace Transformers
- FinBERT
- scikit-learn
- statsmodels
- matplotlib
- seaborn
- Jupyter Notebook

09 Sentiment Distribution

DAILY NET SENTIMENT DISTRIBUTION (S_t)



10 Repository & Files

github.com/cuser7/semicon-news-vol-dataset
INCLUDED ARTIFACTS
CSV LSEG_headlines_clean.csv
IPYNB replication_notebook.ipynb
CSV variable_dictionary.csv
MD README.md